

# Deep Generative Models

## Lecture 5

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# Recap of Previous Lecture

## Training

1. Sample  $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ,  $\epsilon \sim p(\epsilon)$ .
2. Reparametrize  $\mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)$ .
3. Compute the ELBO:

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) \approx \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})).$$

4. Update  $\phi$ ,  $\theta$  via stochastic gradient ascent.

## Sampling

1. Sample  $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$ .
2. Sample  $\mathbf{x} \sim p_{\theta}(\mathbf{x}|\mathbf{z})$ .

**Note:** The encoder  $q_{\phi}(\mathbf{z}|\mathbf{x})$  isn't needed during generation.

# Recap of Previous Lecture

## Theorem

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \parallel p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}].$$

## Revisiting the ELBO

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi,\theta}(\mathbf{x}_i) &= \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} \\ &\quad - \underbrace{\mathbb{I}_q[\mathbf{x}, \mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z}))}_{\text{Marginal KL}} \end{aligned}$$

## Optimal VAE Prior

$$D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) = 0 \Leftrightarrow p(\mathbf{z}) = q_{\text{agg},\phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i).$$

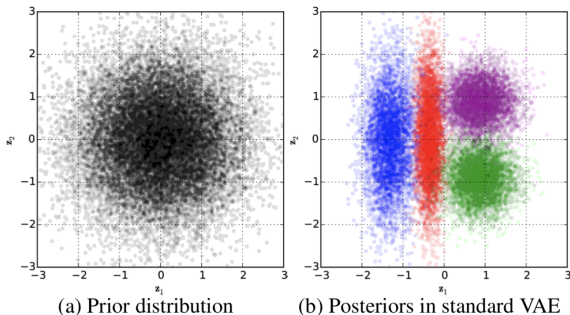
Thus, the optimal prior distribution  $p(\mathbf{z})$  is the aggregated

variational posterior  $q_{\text{agg},\phi}(\mathbf{z})$ .

Hoffman M. D., Johnson M.agg, ELBO: Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound, 2016

# Recap of Previous Lecture

- ▶ **Prior mismatch:** The unimodal encoder  $q_\phi(\mathbf{z}|\mathbf{x})$  yields  $q_{\text{agg},\phi}(\mathbf{z})$  that often does not match the Gaussian prior  $p(\mathbf{z})$ .
- ▶ **Blurriness from averaging:** With Gaussian decoder, ELBO minimization gives  $\mu^*(\mathbf{z}) = \mathbb{E}_{q_\phi(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$ . If distinct inputs  $\mathbf{x} \neq \mathbf{x}'$  map to *overlapping* latent regions, the decoder averages over unrelated data.



# Recap of Previous Lecture

## Assumptions

- ▶ Let  $c \sim \text{Cat}(\boldsymbol{\pi})$ , where

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^K \pi_k = 1.$$

- ▶ Suppose the VAE adopts a discrete latent variable  $c$  with prior  $p(c) = \text{Uniform}\{1, \dots, K\}$ .

## ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x}|c) - D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \| p(c)) \rightarrow \max_{\phi, \theta}.$$

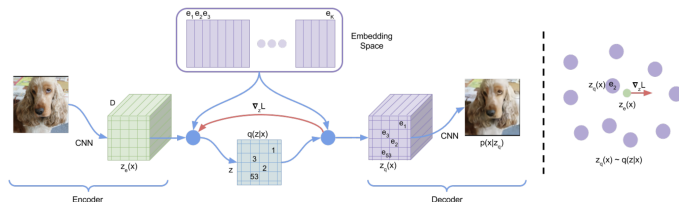
$$D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \| p(c)) = -H(q_{\phi}(c|\mathbf{x})) + \log K.$$

## Quantized Representation

Define the codebook (dictionary) space  $\{\mathbf{e}_k\}_{k=1}^K$  with  $\mathbf{e}_k \in \mathbb{R}^L$  and  $K$  the number of codebook entries.

$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg \min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

# Recap of Previous Lecture



## Deterministic Variational Posterior

$$q_\phi(c = k^* | \mathbf{x}) = \begin{cases} 1, & \text{for } k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

## ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_\phi(c|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{e}_c) - \log K = \log p_\theta(\mathbf{x}|\mathbf{z}_q) - \log K.$$

## Straight-Through Gradient Estimator

$$\frac{\partial \log p(\mathbf{x}|\mathbf{z}_q, \theta)}{\partial \phi} = \frac{\partial \log p_\theta(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \phi} \approx \frac{\partial \log p_\theta(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \phi}$$

# Recap of Previous Lecture

## Likelihood-Free Learning

- ▶ Likelihood isn't always a suitable metric for evaluating generative models.
- ▶ Sometimes, the likelihood function can't even be computed exactly.

Imagine we have two sets of samples:

- ▶  $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$  – real samples;
- ▶  $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\theta}(\mathbf{x})$  – generated (fake) samples.  
 $p(y=1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y=0|\mathbf{x}) = P(\mathbf{x} \sim p_{\theta}(\mathbf{x}))$

## Assumption

The generative model  $p_{\theta}(\mathbf{x})$  matches  $p_{\text{data}}(\mathbf{x})$  if a discriminative model  $p(y|\mathbf{x})$  can't distinguish between them — that is, if

$p(y=1|\mathbf{x}) = 0.5$  for every  $\mathbf{x}$ .

- ▶ **Generator:** a generative model  $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$  that produces more realistic samples.
- ▶ **Discriminator:** a classifier  $D_{\phi}(\mathbf{x}) \in [0, 1]$  distinguishing real from generated samples.

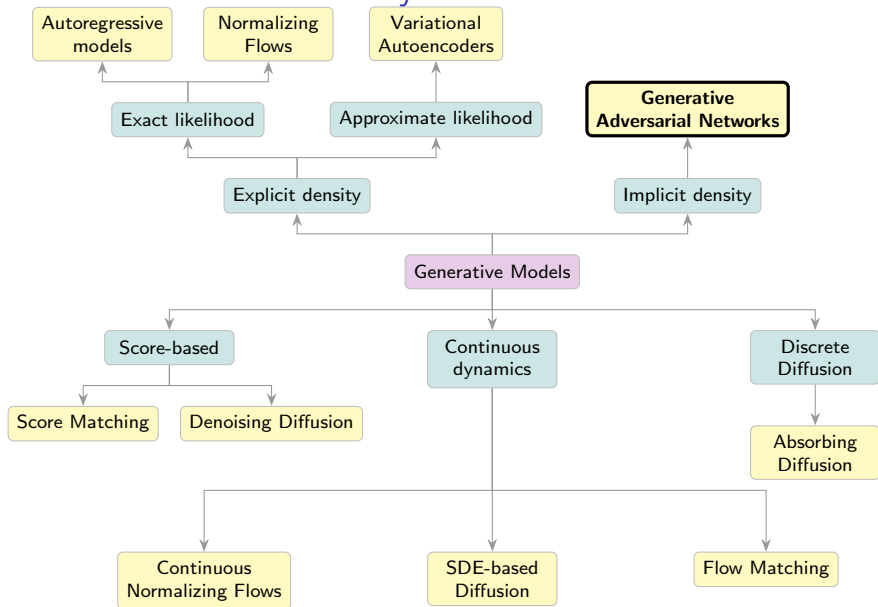
# Outline

1. Generative Adversarial Networks (GAN)
2. Wasserstein Distance
3. Wasserstein GAN (WGAN)
4. Evaluation of Likelihood-Free Models
  - Frechet Inception Distance (FID)
  - Precision-Recall
  - CLIP Score
  - Human Eval

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# Generative Models Taxonomy



# GAN Optimality

## Theorem

The minimax game

$$\min_G \max_D \underbrace{\left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]}_{V(G,D)}$$

achieves its global optimum when  $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$ , and  $D^*(\mathbf{x}) = 0.5$ .

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## Proof (Fixed $G$ )

$$V(G, D) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x}))$$

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$$\frac{dy(D)}{dD} = \frac{p_{\text{data}}(\mathbf{x})}{D(\mathbf{x})} - \frac{p_{\theta}(\mathbf{x})}{1 - D(\mathbf{x})} = 0 \quad \Rightarrow \quad D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})}$$

# GAN Optimality

Proof Continued (Fixed  $D = D^*$ )

$$V(G, D^*) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log \left( \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log \left( \frac{p_{\theta}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right)$$

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## Jensen-Shannon Divergence (Symmetric KL Divergence)

$$D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) = \frac{1}{2} [D_{\text{KL}}(p_{\text{data}}(\mathbf{x}) \parallel \star) + D_{\text{KL}}(p_{\theta}(\mathbf{x}) \parallel \star)]$$

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This can be regarded as a proper distance metric!

$$V(G^*, D^*) = -2 \log 2, \quad p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x}), \quad D^*(\mathbf{x}) = 0.5.$$

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The following minimax game

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## Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

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## Reality

- ▶ Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- ▶ Generator and discriminator losses typically oscillate during GAN training.

# GAN Training

Assume both generator and discriminator are parametric models:  
 $D_\phi(\mathbf{x})$  and  $\mathbf{G}_\theta(\mathbf{z})$ .

## Objective

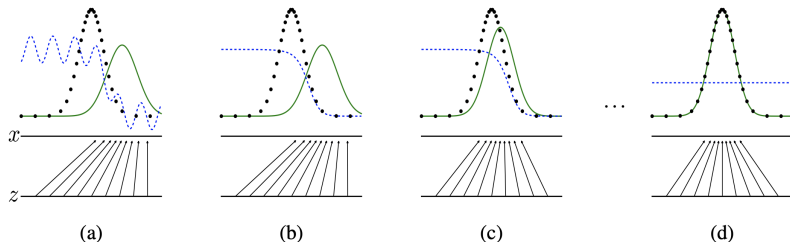
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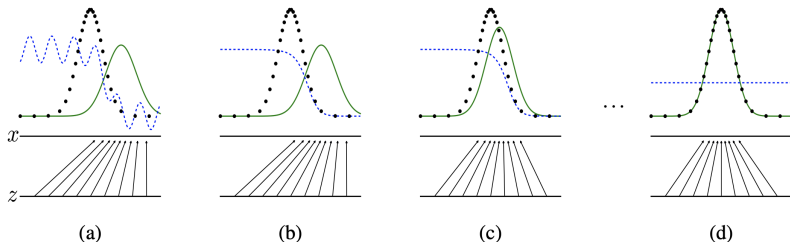


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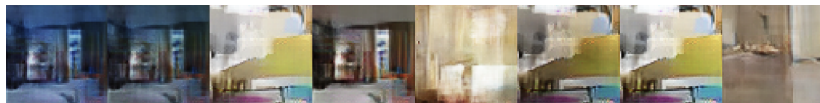
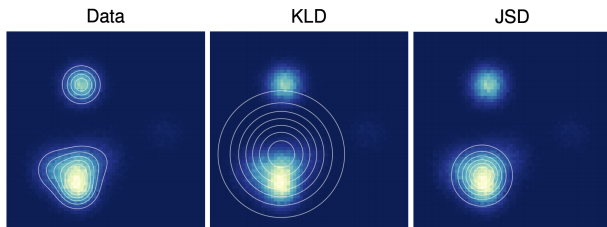
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- ▶  $\mathbf{z} \sim p(\mathbf{z})$  is a latent variable.
- ▶  $p_\theta(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_\theta(\mathbf{z}))$  serves as a deterministic decoder (like normalizing flows).
- ▶ There is no encoder present.

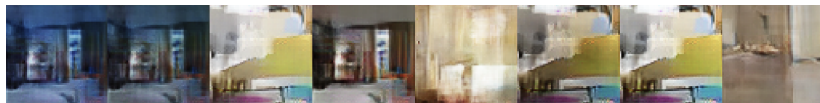
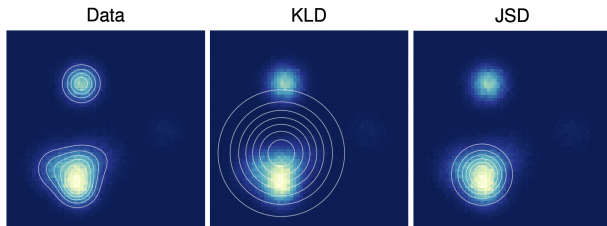
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Numerous methods have been proposed to tackle mode collapse: changing architectures, adding regularization terms, injecting noise.

*Goodfellow I. J. et al. Generative Adversarial Networks, 2014*

*Metz L. et al. Unrolled Generative Adversarial Networks, 2016*

# Jensen-Shannon vs Kullback-Leibler Divergences

- ▶  $p_{\text{data}}(\mathbf{x})$  is a fixed mixture of two Gaussians.
- ▶  $p(\mathbf{x}|\mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$ .

## Mode Covering vs. Mode Seeking

$$D_{\text{KL}}(\pi \parallel p) = \int \pi(\mathbf{x}) \log \frac{\pi(\mathbf{x})}{p(\mathbf{x})} d\mathbf{x}, \quad D_{\text{KL}}(p \parallel \pi) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$$

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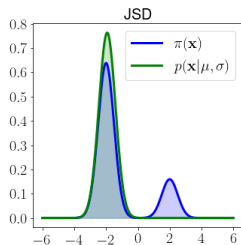
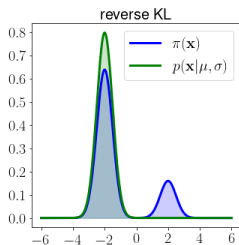
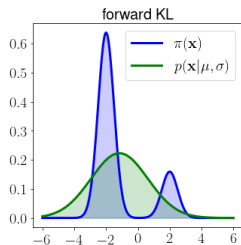
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# Theoretical Results

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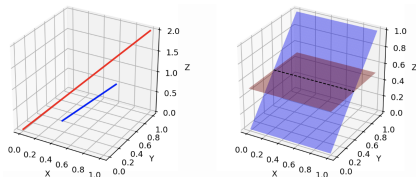
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Weng L. *From GAN to WGAN*, 2019

Arjovsky M., Bottou L. *Towards Principled Methods for Training Generative Adversarial Networks*, 2017

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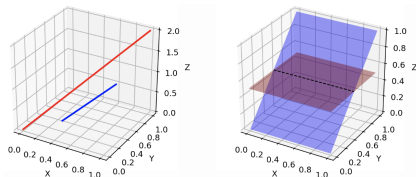
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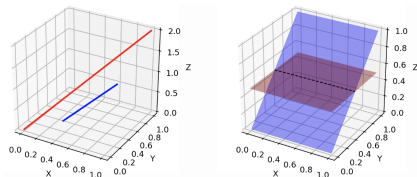
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- ▶ If  $p_{\text{data}}(\mathbf{x})$  and  $p_{\theta}(\mathbf{x})$  are disjoint, a smooth optimal discriminator can exist!
- ▶ For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \parallel p_{\theta}) = D_{\text{KL}}(p_{\theta} \parallel p_{\text{data}}) = \infty, \quad D_{\text{JS}}(p_{\text{data}} \parallel p_{\theta}) = \log 2$$

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Weng L. *From GAN to WGAN*, 2019

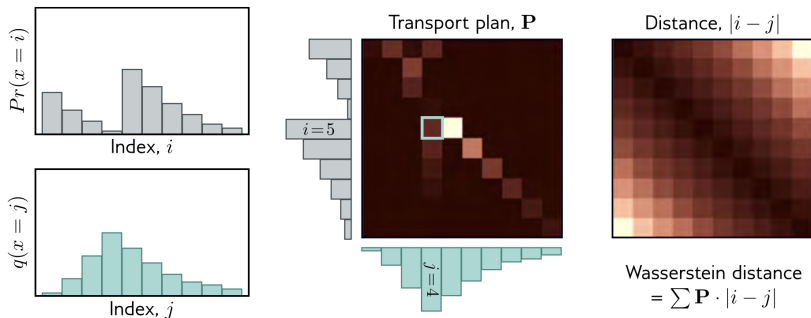
Arjovsky M., Bottou L. *Towards Principled Methods for Training Generative Adversarial Networks*, 2017

# Wasserstein Distance (Discrete)

Also known as the **Earth Mover's Distance**.

## Optimal Transport Formulation

The minimum cost of moving and transforming a pile of “dirt” shaped like one probability distribution to match another.



# Wasserstein Distance (Continuous)

$$W(\pi \| p) = \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| = \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2$$

- ▶  $\gamma(\mathbf{x}_1, \mathbf{x}_2)$  is the transport plan: the amount of “dirt” assigned from  $\mathbf{x}_1$  to  $\mathbf{x}_2$ .

$$\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2); \quad \int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1).$$

- ▶  $\Gamma(\pi, p)$  denotes the set of all joint distributions  $\gamma(\mathbf{x}_1, \mathbf{x}_2)$  with marginals  $\pi$  and  $p$ .
- ▶  $\gamma(\mathbf{x}_1, \mathbf{x}_2)$  is the mass,  $\|\mathbf{x}_1 - \mathbf{x}_2\|$  is the distance.

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## Wasserstein Metric

$$W_s(\pi, p) = \inf_{\gamma \in \Gamma(\pi, p)} \left( \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

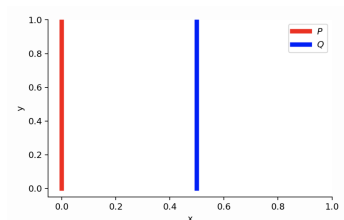
In our setting,  $W(\pi \| p) = W_1(\pi, p)$ , which is the transport cost formulation.

# Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1])$$

$$p_{\theta}(x, y) = (\theta, U[0, 1])$$



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Weng L. *From GAN to WGAN*, 2019

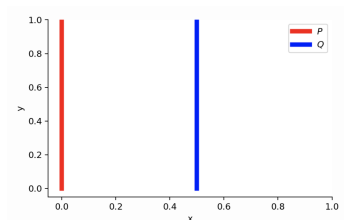
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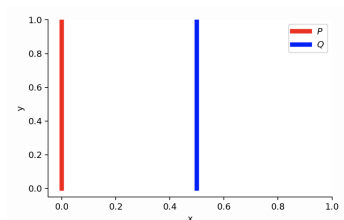
$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

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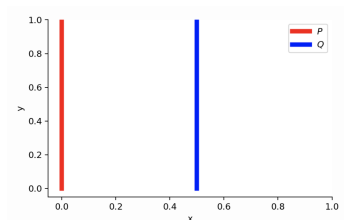
$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int_{U[0,1]} 1 \log \frac{1}{0} dy = \infty = D_{\text{KL}}(p_{\theta} \| p_{\text{data}})$$

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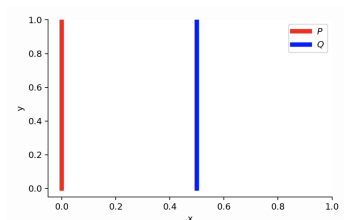
$$D_{\text{JS}}(p_{\text{data}} \| p_{\theta}) = \frac{1}{2} \left( \int_{U[0,1]} 1 \log \frac{1}{1/2} dy + \int_{U[0,1]} 1 \log \frac{1}{1/2} dy \right) = \log 2$$

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$$W(p_{\text{data}} \| p_{\theta}) = |\theta|$$

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Arjovsky M., Chintala S., Bottou L. *Wasserstein GAN*, 2017

# Wasserstein Distance vs KL vs JSD

## Theorem 1

Let  $\mathbf{G}_\theta(\mathbf{z})$  be (almost) any feedforward neural network, and  $p(\mathbf{z})$  a prior over  $\mathbf{z}$  such that  $\mathbb{E}_{p(\mathbf{z})} \|\mathbf{z}\| < \infty$ . Then  $W(p_{\text{data}} \| p_\theta)$  is continuous everywhere and differentiable almost everywhere.

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## Theorem 2

Let  $\pi$  be a distribution on a compact space  $\mathcal{X}$  and let  $\{p_t\}_{t=1}^\infty$  be a sequence of distributions on  $\mathcal{X}$ .

$$D_{\text{KL}}(\pi \| p_t) \rightarrow 0 \quad (\text{or } D_{\text{KL}}(p_t \| \pi) \rightarrow 0) \quad (1)$$

$$D_{\text{JS}}(\pi \| p_t) \rightarrow 0 \quad (2)$$

$$W(\pi \| p_t) \rightarrow 0 \quad (3)$$

As  $t \rightarrow \infty$ ,  $(1) \Rightarrow (2)$ , and  $(2) \Rightarrow (3)$ . That is, convergence in Wasserstein distance is a weaker condition than convergence in JSD, which in turn is weaker than convergence in KL.

# Outline

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# Wasserstein GAN

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$$W(\pi \| p) = \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| = \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2$$

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## Theorem (Kantorovich-Rubinstein Duality)

$$W(\pi \| p) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[ \mathbb{E}_{\pi(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p(\mathbf{x})} f(\mathbf{x}) \right]$$

where  $f : \mathbb{R}^m \rightarrow \mathbb{R}$  is  $K$ -Lipschitz ( $\|f\|_L \leq K$ ):

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq K \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}.$$

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We can thus estimate  $W(\pi \| p)$  using only samples and a function  $f$ .

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- ▶ We must ensure that  $f$  is  $K$ -Lipschitz continuous.
- ▶ Let  $f_{\phi}(\mathbf{x})$  be a feedforward neural network parameterized by  $\phi$ .
- ▶ If the weights  $\phi$  are restricted to a compact set  $\Phi$ , then  $f_{\phi}$  is  $K$ -Lipschitz.

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- ▶ Clamp weights within the box  $\Phi = [-c, c]^d$  (e.g.  $c = 0.01$ ) after each update.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\theta}) &= \max_{\|f\|_L \leq K} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right] \geq \\ &\geq \max_{\phi \in \Phi} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f_{\phi}(\mathbf{x}) \right] \end{aligned}$$

# Wasserstein GAN

## Standard GAN Objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})))$$

## WGAN Objective

$$\min_{\theta} W(p_{\text{data}} \| p_{\theta}) \approx \min_{\theta} \max_{\phi \in \Phi} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})) \right]$$

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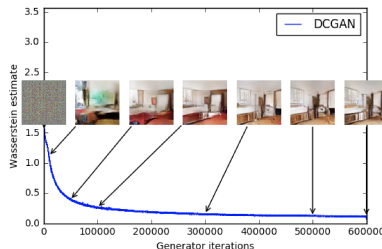
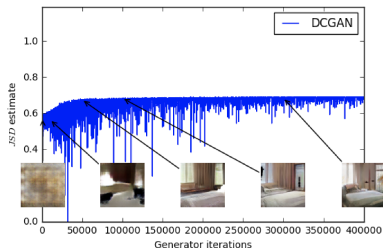
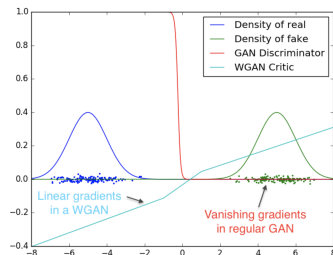
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- ▶ The discriminator  $D$  is replaced by function  $f$ : in WGAN, it is known as the **critic**, which is *not* a classifier.
- ▶ “*Weight clipping is a clearly terrible way to enforce a Lipschitz constraint.*”
  - ▶ If  $c$  is large, optimizing the critic is hard.
  - ▶ If  $c$  is small, gradients may vanish.

# Wasserstein GAN

- ▶ WGAN provides nonzero gradients even if distributions' supports are disjoint.
- ▶  $D_{JS}(p_{\text{data}} \| p_{\theta})$  is poorly correlated with sample quality and remains near its maximum value  $\log 2 \approx 0.69$ .
- ▶  $W(p_{\text{data}} \| p_{\theta})$  is tightly correlated with quality.



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# Evaluation of Likelihood-Free Models

## Likelihood-Based Models

- ▶ **Train:** fit the model.
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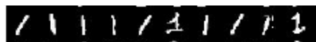


Low sharpness

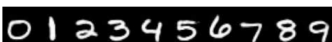


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## Theorem

If  $\pi(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\pi, \boldsymbol{\Sigma}_\pi)$ ,  $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$ , then

$$W_2^2(\pi \| p) = \|\boldsymbol{\mu}_\pi - \boldsymbol{\mu}_p\|^2 + \text{tr} \left[ \boldsymbol{\Sigma}_\pi + \boldsymbol{\Sigma}_p - 2 \left( \boldsymbol{\Sigma}_\pi^{1/2} \boldsymbol{\Sigma}_p \boldsymbol{\Sigma}_\pi^{1/2} \right)^{1/2} \right]$$

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## Frechet Inception Distance

$$\text{FID}(p_{\text{data}}, p_\theta) = W_2^2(p_{\text{data}} \| p_\theta)$$

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Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017

# Frechet Inception Distance (FID)

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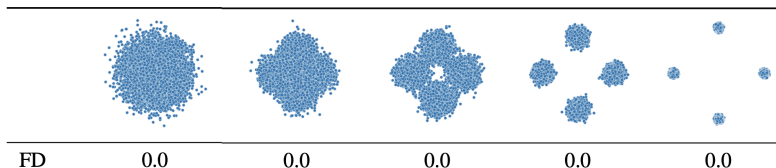
- ▶ FID is computed in the latent space  $\mathbf{z}$ .
- ▶ We use a pretrained image embedder to get latent representations  $\mathbf{z} = \mathbf{f}(\mathbf{x})$ .
- ▶  $\mu_{\text{data}}$ ,  $\Sigma_{\text{data}}$  and  $\mu_{\theta}$ ,  $\Sigma_{\theta}$  are statistics of latent representations for samples from  $p_{\text{data}}(\mathbf{x})$  and  $p_{\theta}(\mathbf{x})$ .

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$$\text{FID}(p(\mathbf{x}), \mathcal{N}(0, \mathbf{I}))$$



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$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\mu_{\text{data}} - \mu_{\theta}\|^2 + \text{tr} \left[ \Sigma_{\text{data}} + \Sigma_{\theta} - 2 \left( \Sigma_{\text{data}}^{1/2} \Sigma_{\theta} \Sigma_{\text{data}}^{1/2} \right)^{1/2} \right]$$

## Drawbacks

- ▶ Depends on the pretrained classification network.
- ▶ Uses the normality assumption.
- ▶ May not correlate with human evaluation.

| Model                  | Model-A | Model-B |
|------------------------|---------|---------|
| FID                    | 21.40   | 18.42   |
| FID <sub>∞</sub>       | 20.16   | 17.19   |
| Human rater preference | 92.5%   | 6.9%    |

# Outline

1. Generative Adversarial Networks (GAN)
2. Wasserstein Distance
3. Wasserstein GAN (WGAN)
4. Evaluation of Likelihood-Free Models
  - Frechet Inception Distance (FID)
  - Precision-Recall
  - CLIP Score
  - Human Eval

# Precision-Recall

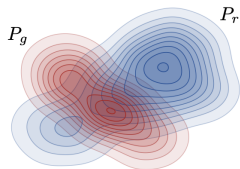
## Desirable Properties for Samples

- ▶ **Sharpness**: generated samples should possess high visual quality.
- ▶ **Diversity**: their variation should match that in the training data.

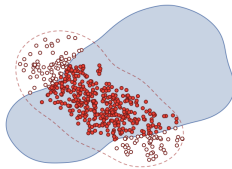
# Precision-Recall

## Desirable Properties for Samples

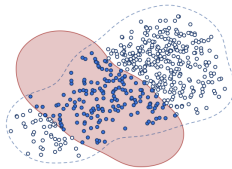
- ▶ **Sharpness**: generated samples should possess high visual quality.
- ▶ **Diversity**: their variation should match that in the training data.



(a) Example distributions



(b) Precision



(c) Recall

- ▶ **Precision** denotes the fraction of generated images that look realistic.
- ▶ **Recall** measures how well the generator covers the training data manifold.

# Precision-Recall

- ▶  $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$  – real samples;
- ▶  $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$  – generated samples.

# Precision-Recall

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Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

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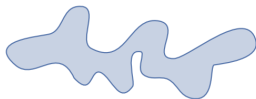
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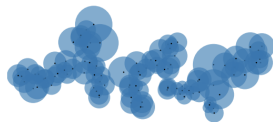
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(a) True manifold



(b) Approx. manifold

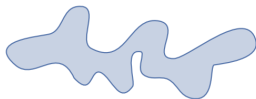
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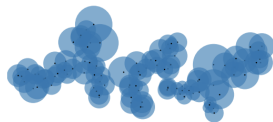
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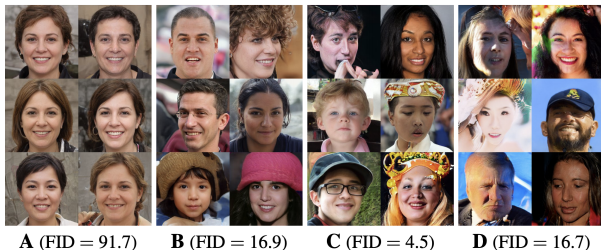
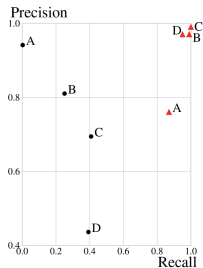
(a) True manifold



(b) Approx. manifold

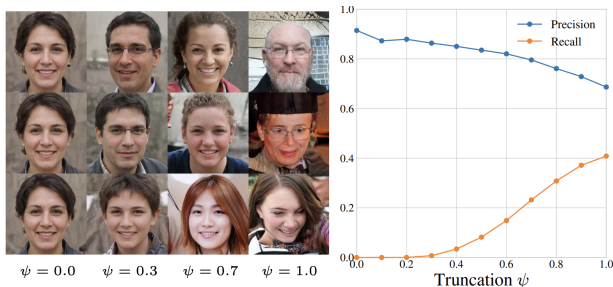
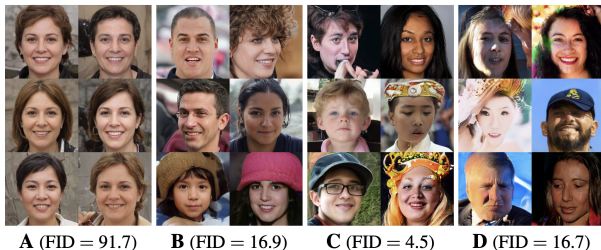
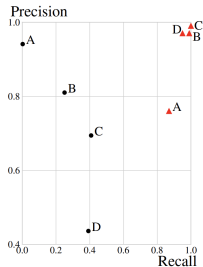
Embed the samples using a pretrained network (as in FID).

# Precision-Recall



Kynkäänniemi T. et al. Improved precision and recall metric for assessing generative models, 2019

# Precision-Recall



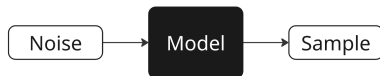
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# Outline

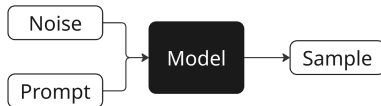
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# CLIP Score

## Unconditional Model

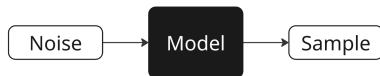


## Conditional Model

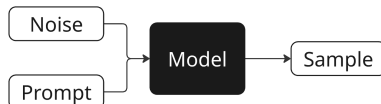


# CLIP Score

## Unconditional Model



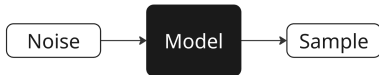
## Conditional Model



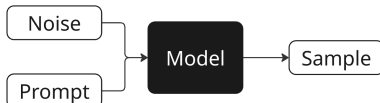
We need a way to measure not only the quality of the generated image, but also how well it's aligned with the prompt.

# CLIP Score

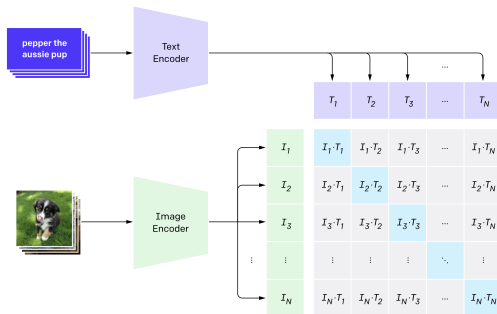
## Unconditional Model



## Conditional Model



We need a way to measure not only the quality of the generated image, but also how well it's aligned with the prompt.



Radford A. et al. *Learning transferable visual models from natural language supervision*, 2021

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# Human Evaluation

- ▶ No automated metric is perfect.
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| Аспект        | Yandex ART 2.0 | Mj 6.1 | Mj 6 | Ideogram | Recraft | Google Imagen3 | Dall-E 3 | FLUX | SBER Kandi3.1 |
|---------------|----------------|--------|------|----------|---------|----------------|----------|------|---------------|
| Релевантность | 0,59           | 0,58   | 0,63 | 0,45     | 0,51    | 0,50           | 0,50     | 0,54 | 0,75          |
| Эстетика      | 0,49           | 0,55   | 0,55 | 0,51     | 0,51    | 0,61           | 0,61     | 0,54 | 0,59          |
| Комплексность | 0,44           | 0,73   | 0,70 | 0,68     | 0,76    | 0,75           | 0,75     | 0,71 | 0,74          |
| Дефектность   | 0,69           | 0,57   | 0,68 | 0,55     | 0,59    | 0,63           | 0,63     | 0,50 | 0,75          |
| Предпочтение  | 0,66           | 0,60   | 0,69 | 0,49     | 0,54    | 0,63           | 0,63     | 0,51 | 0,84          |

## Summary

- ▶ GANs, in theory, optimize the Jensen-Shannon divergence.
- ▶ Wasserstein distance works in the case of disjoint data and model distributions (unlike the KL and JS divergences).
- ▶ Wasserstein GAN uses Kantorovich-Rubinstein duality to enable Monte Carlo estimation of the Wasserstein distance. It enforces the Lipschitz condition on the critic through weight clipping.
- ▶ FID is the most popular metric for evaluating implicit generative models.
- ▶ Precision-recall allows for choosing a model that balances sample quality and diversity.
- ▶ The CLIP score is widely used to measure text-to-image alignment.
- ▶ The gold standard for evaluating generated image quality is human assessment.